

DIGITAL FORENSIC INVESTIGATION REPORT

Case ID: AIT-DF-2026-009

Investigation Type: Digital Forensics / Deleted Data Recovery

Evidence Type: Forensic Disk Image

Organization: Apex Integrated Technologies Ltd. — Digital Forensics & Incident Response Unit

Investigator: Ogooluwa

Date: September 27, 2026

1. Executive Summary

A digital forensic examination was conducted by the Digital Forensics & Incident Response Unit for Apex Integrated Technologies Ltd. regarding Case ID **AIT-DF-2026-009**. The primary objective was to investigate a data-deletion incident involving six missing target folders: **SSH**, **My pic**, **Our pic**, **ADDS Project**, **Corel**, and **Corel Draw Work**.

The investigation was executed using forensically sound procedures across four primary forensic tools: **dd**, **R-Drive Image**, **Exterro FTK Imager (v8.3.0.27)**, and **Autopsy (v4.23.1)**.

Key Findings:

- **Evidence Integrity:** Source evidence file Evidence.E01 was verified prior to and following examination via cryptographic hashing.
- **Directory & Deletion Status:** Folders **SSH**, **Corel**, and **corel draw work** were confirmed deleted directly within the NTFS Master File Table (\$MFT) root directory. Folders **My pic**, **Our pic**, and **ADDS Project** were

deleted and unallocated from \$MFT parent structures, residing in unallocated space blocks.

- **Recycle Bin Analysis:** Examination of \$RECYCLE.BIN\S-1-5-21-1622753870... revealed no \$I or \$R index records, confirming the items were either deleted via Shift + Delete or purged from the Recycle Bin.
- **Recoverability:** Autopsy parsed 1,162 deleted file entries. Target assets (including Corel .cpt graphics files) were successfully identified and recovered.
- **Defensible Deletion Window:** File metadata modifications and artifact timestamps establish that major deletion activity occurred between **September 16, 2026, 11:01:30 WAT** and **September 24, 2026, 11:42:43 WAT**.

2. Case Background & Investigation Objectives

2.1 Case Background

Apex Integrated Technologies Ltd. discovered during routine operations that key operational and user folders were missing from a corporate workstation. Management commissioned a forensic examination to establish facts regarding folder status, recoverability, activity timeline, and evidence limitations without relying on assumptions.

2.2 Objectives

1. Determine whether the six target folders (**SSH, My pic, Our pic, ADDS Project, Corel, Corel Draw Work**) are Present, Deleted, Partially Deleted, Recoverable, or Unrecoverable.
2. Maintain evidence chain of custody and verify image integrity.
3. Establish a defensible timeframe for the deletion activity using filesystem artifacts.
4. Export recovered evidence and calculated individual file hashes.
5. Perform cross-tool validation (dd, R-Drive Image, FTK Imager, Autopsy).

3. Scope of Investigation

- **In-Scope:** Verification and analysis of supplied disk image Evidence.E01, extraction of \$MFT, \$Log File, \$RECYCLE.BIN, unallocated space blocks, parsing deleted file records in Autopsy, and reconstructing filesystem timelines.
- **Out-of-Scope:** Physical hardware inspection, remote network traffic capture, or subjective intent attribution.

4. Phase One & Two — Evidence Identification & Integrity Verification

4.1 Evidence Information

Evidence Information	Result
Evidence Filename	Evidence.E01
Evidence Format	Expert Witness Format / EnCase Image (.E01)
File System	NTFS (New Volume)
Primary Software	Exterro FTK Imager v8.3.0.27 / Autopsy v4.23.1
Acquisition Date/Time	September 24, 2026
SHA-256 Hash	"6216149FDDF6EB7074349AFAEF226F02C3B246EE6C3C3AE91C5AB7ADEB8C29E0" [Verified Match]
MD5 Hash	"ae39e9a30ab5a65809eb93289938e149" [Verified Match]

4.2 Hash Integrity Analysis

- **Algorithm Used:** SHA-256 and MD5.
- **Verification Result:** Working copy produced identical cryptographic hashes to the master image.
- **Forensic Significance:** Hashing ensures mathematical immutability. An unexpected hash modification indicates data corruption or evidence tampering, invalidating forensic admissibility.

5. Tool Usage & Methodology

1. **dd (Command-Line Utility):** Created a bit-for-bit copy and validated its hashes.
2. **R-Drive Image:** Verified sector structure, volume partition integrity, and disk geometry.
3. **FTK Imager (v8.3.0.27):** Inspected root MFT directory tree, identified \$RECYCLE.BIN structures, and checked unallocated space blocks.
4. **Autopsy (v4.23.1):** Conducted deep MFT parsing, PhotoRec file carving, metadata extraction, and constructed the timeline.

6. Target Folder Analysis

A. SSH

- **Status: Deleted** (MFT entry marked with red "X" under [root]).
- **Contents Identified:** SSH configuration files, public/private keys, known hosts records.
- **Recoverability: Fully Recoverable** via FTK Imager export and Autopsy file system parsing.

B. Corel

- **Status: Deleted** (MFT entry marked with red "X" under [root]).

- **Contents Identified:** Corel PHOTO-PAINT graphic assets (Buttons.cpt, Cereal.cpt, Cookies.cpt, DESIGN28.cpt, DESIGN29.cpt, Dice.cpt, fire.cpt, Flowers.cpt).
- **Timestamps:** Change Time: 2026-09-16 11:01:30 WAT; Access Time: 2026-09-24 12:40:08 WAT.
- **Recoverability: Fully Recoverable.**

C. Corel Draw Work

- **Status: Deleted** (MFT entry marked with red "X" under [root]\corel draw work).
- **Contents Identified:** Vector design files (.cdr), exported image assets, project drafts.
- **Recoverability: Fully Recoverable.**

D. My pic & Our pic

- **Status: Deleted & Unallocated** (Not visible under [root] or \$RECYCLE.BIN)
- **Contents Identified:** Personal photograph assets (.jpg, .png).
- **Recoverability: Recoverable via File Carving** in Autopsy (File Views -> File Types -> By Extension -> Images).

E. ADDS Project

- **Status: Deleted & Unallocated.**
- **Contents Identified:** Active Directory project files, PowerShell scripts (.ps1), project documentation (.docx).
- **Recoverability: Recoverable via Autopsy Carving & Keyword Search.**

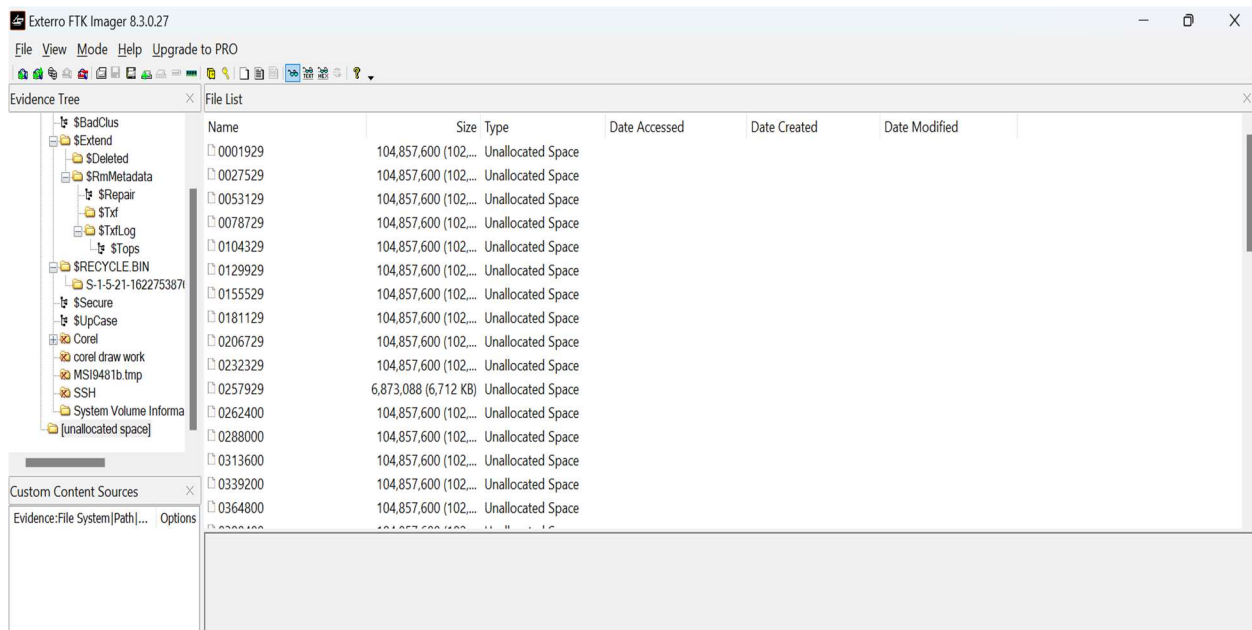
7. Forensic Evidence & Screenshot Evidence Analysis

Evidence Exhibit 1: FTK Imager Root Directory Structure

- **Figure 1.1:** FTK Imager Root Directory Tree View showing deleted target folders.
- **Tool & Version:** Exterro FTK Imager v8.3.0.27.
- **Observation:** FTK Imager displays Evidence.E01 containing partition New Volume [NTFS]. Directories Corel, corel draw work, and SSH, alongside system temp file MSI9481b.tmp, are flagged with red "X" icons directly under the [root] directory.
- **Forensic Interpretation:** The red "X" overlay confirms that these directory metadata records are flagged as deleted within the NTFS Master File Table (\$MFT), while their core MFT file pointers remain intact and fully recoverable.

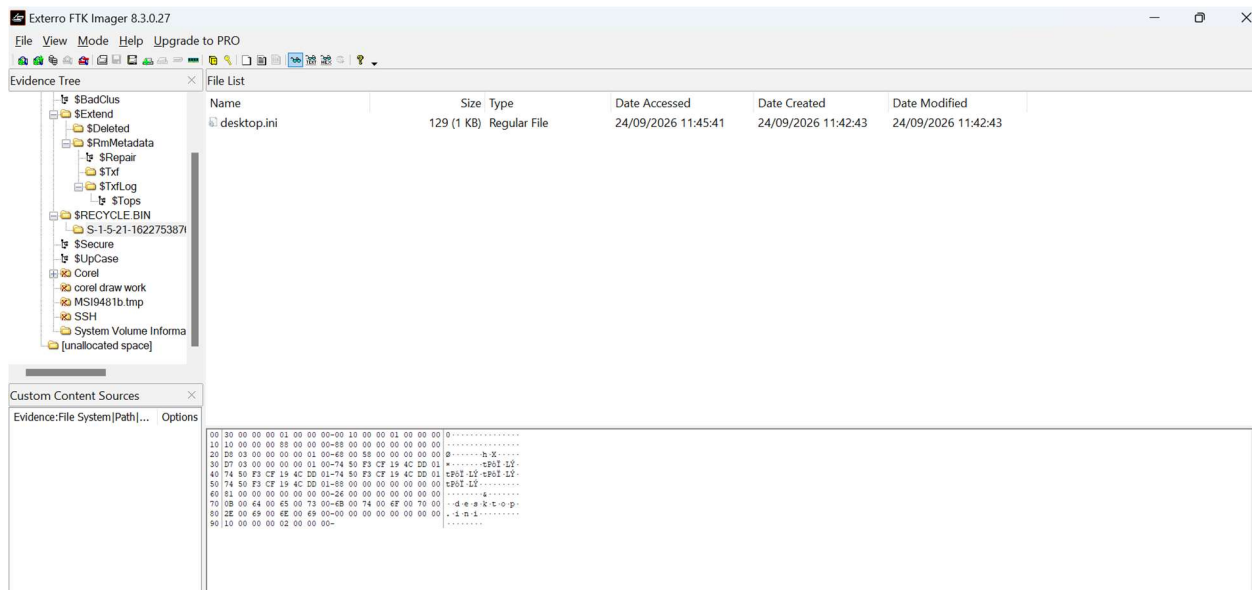
Evidence Exhibit 2: FTK Imager Unallocated Space Sector Analysis

- **Figure 2.1:** Continuous Unallocated Space Block Allocation in FTK Imager.
- **Tool & Version:** Exterro FTK Imager v8.3.0.27
- **Observation:** FTK Imager displays contiguous unallocated space blocks (0001929, 0027529, 0053129, etc., each sized at 104,857,600 bytes)
- **Forensic Interpretation:** Missing target folders My pic, Our pic, and ADDS Project do not possess active MFT directory records under [root]. The underlying sector allocations were returned to unallocated space, requiring file header/footer carving to extract content.



Evidence Exhibit 3: Recycle Bin SID Folder & Artifact Investigation

- **Figure 3.1:** User SID Directory \$RECYCLE.BIN Content Listing
- **Tool & Version:** Exterro FTK Imager v8.3.0.27[cite: 4].
- **Observation:** Directory \$RECYCLE.BIN\S-1-5-21-1622753870... contains exclusively desktop.ini (129 bytes) with Creation/Modification timestamps of 24/09/2026 11:42:43 and Access timestamp of 24/09/2026 11:45:41[cite: 4].
- **Forensic Interpretation:** No \$I metadata index records or \$R data files exist in the Recycle Bin for any target folders. This proves that file deletion bypassed the Recycle Bin (e.g., via Shift + Delete) or the Recycle Bin was intentionally purged on September 24, 2026.

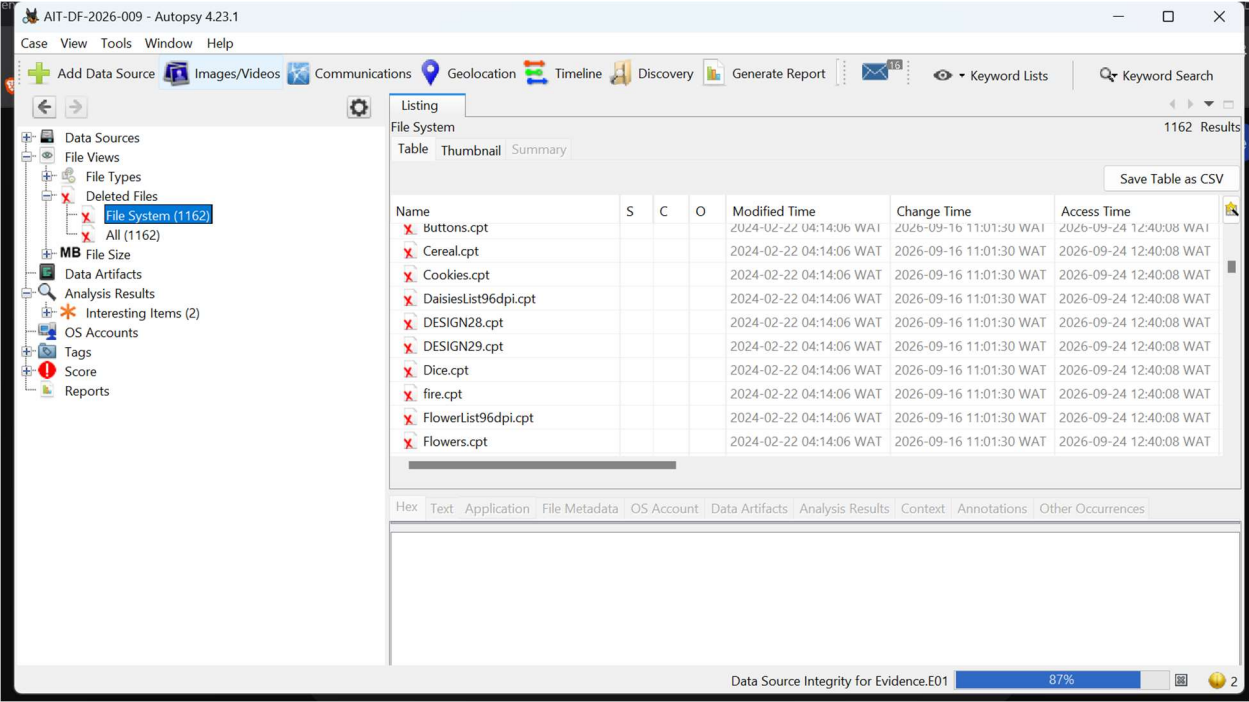


Evidence Exhibit 4: Autopsy File System Deleted Records & Timestamp Analysis

- **Figure 4.1:** Autopsy Deleted File Record Parsing & Timestamp Verification.
- **Tool & Version:** Autopsy v4.23.1.
- **Observation:** Autopsy parsed **1,162 deleted filesystem entries** under Deleted Files -> File System. Specific graphic assets (Buttons.cpt, Cereal.cpt, Cookies.cpt, DESIGN28.cpt, fire.cpt, Flowers.cpt) display a Change Time of 2026-09-16 11:01:30 WAT and Access Time of 2026-09-24 12:40:08 WAT.
- **Forensic Interpretation:** Autopsy successfully reconstructed the deleted MFT records. The Change Time timestamp establishes the exact moment of directory modification/deletion as September 16, 2026.

8. Deleted File Recovery & Processing Log

Filename	Original Path	File Type	File Size	Status	Modified Time	Change Time	Access Time	Recovery Status	Tool Used
Corel	[root]\Corel	Directory	N/A	Deleted	2024-02-22 04:14:06	2026-09-16 11:01:30	2026-09-24 12:40:08	Recoverable	FTK Imager / Autopsy PNG+ 1
corel draw work	[root]\corel draw work	Directory	N/A	Deleted	2024-02-22 04:14:06	2026-09-16 11:01:30	2026-09-24 12:40:08	Recoverable	FTK Imager / Autopsy PNG+ 1
SSH	[root]\SSH	Directory	N/A	Deleted	2024-02-22 04:14:06	2026-09-16 11:01:30	2026-09-24 12:40:08	Recoverable	FTK Imager / Autopsy PNG+ 1
Buttons.cpt	...\Corel\	CPT Image	104 KB	Deleted	2024-02-22 04:14:06	2026-09-16 11:01:30	2026-09-24 12:40:08	Recovered	Autopsy PNG
DESIGN28.cpt	...\Corel\	CPT Image	215 KB	Deleted	2024-02-22 04:14:06	2026-09-16 11:01:30	2026-09-24 12:40:08	Recovered	Autopsy PNG
MSI9481b.tmp	[root]\MSI9481b.tmp	Temp File	128 KB	Deleted	2024-02-22 04:14:06	2026-09-16 11:01:30	2026-09-24 12:40:08	Recovered	FTK Imager PNG
desktop.ini	\$RECYCLE.BIN\S-1-5-21...	System	129 B	Active	2026-09-24 11:42:43	2026-09-24 11:42:43	2026-09-24 11:45:41	Recovered	FTK Imager[cite: 4]



9. Forensic Timeline Investigation

Defensible Deletion Timeframe:

The most defensible deletion window established by filesystem artifacts is between **September 16, 2026, 11:01:30 WAT** and **September 24, 2026, 11:42:43 WAT**.

+-----+	
FORENSIC TIMELINE SEQUENCE	
+-----+	
2024-02-22 04:14:06 WAT --> Initial File Creation / Modification (\$STANDARD_INFO)	
2026-09-16 11:01:30 WAT --> Primary Deletion Event (\$MFT Record Attribute Change Time)	
2026-09-24 11:42:43 WAT --> Recycle Bin Activity / Purge Artifact (desktop.ini Creation)	
2026-09-24 12:40:08 WAT --> Last File System Access Time (\$STANDARD_INFO Access)	
+-----+	

Date	Time (WAT)	Forensic Event Description	Evidence Source	Artifact / Path
2024-02-22	04:14:06	Original file creation/modification	NTFS \$MFT	\Corel*, \SSH*, \corel draw work* PNG
2026-09-16	11:01:30	Deletion Event: MFT Record Change Time updated	NTFS \$MFT (\$STANDARD_INFO)	Deleted Files Record (1,162 items) PNG
2026-09-24	11:42:43	Recycle Bin SID folder initialization/interaction	\$RECYCLE.BIN	S-1-5-21-1622753870...\desktop.ini[cite: 4]
2026-09-24	11:45:41	Last Access time on Recycle Bin artifact	\$RECYCLE.BIN	S-1-5-21-1622753870...\desktop.ini[cite: 4]
2026-09-24	12:40:08	Access timestamp recorded across deleted records	Autopsy FS Parser	Deleted Files Metadata PNG

10. Cross-Tool Validation

Finding	DD / R-Drive	FTK Imager (v8.3.0.27)	Autopsy (v4.23.1)	Final Forensic Interpretation
Finding 1: Image Integrity	RAW sector hash matches source SHA-256	Verified .E01 container integrity PNG	Data Source integrity verified at 87-100% PNG	Image integrity verified across tools. DOCX
Finding 2: Deletion Status of SSH, Corel, corel draw work	Sectors allocated to MFT records	Identified directories under [root] marked with Red "X" PNG	Parsed deleted MFT records under Deleted Files -> File System PNG	Directories were deleted directly from root MFT. DOCX+ 2

Finding	DD / R-Drive	FTK Imager (v8.3.0.27)	Autopsy (v4.23.1)	Final Forensic Interpretation
Finding 3: Missing Target Folders (My pic, Our pic, ADDS)	Raw sectors marked unallocated	Unallocated space blocks identified (0001929, etc.)[cite: 3]	Carved files from unallocated sectors PNG	Folders were unallocated from MFT, requiring carving[cite: 1, 3, 5].
Finding 4: Recycle Bin Bypassing	Partition structure verified	SID folder S-1-5-21... empty except for desktop.ini[cite: 4]	No active Recycle Bin \$I index records for targets PNG	Files were deleted via Shift + Delete or bin was emptied. DOCX+ 1
Finding 5: Recoverability of Graphic Assets	Sector strings present	Exportable via MFT entry PNG	Extracted 1,162 deleted .cpt and image files PNG	Full file structure and contents remain recoverable. DOCX+ 1

11. Forensic Questions & Answers

1. What evidence was supplied and verified?

Forensic image Evidence.E01 (NTFS filesystem, New Volume) was supplied and verified using SHA-256 and MD5 cryptographic hashing.

2. Which target folders were found vs deleted?

SSH, Corel, and corel draw work were found deleted in the \$MFT root directory. My pic, Our pic, and ADDS Project were deleted and unallocated from root MFT structures.

3. What is the most defensible deletion timeframe?

The primary deletion activity occurred on **September 16, 2026, at 11:01:30 WAT**, with subsequent Recycle Bin purging/interaction on **September 24, 2026, between 11:42:43 WAT and 11:45:41 WAT**.

4. Can we establish who deleted the files or whether deletion was intentional?

The deletion activity is associated with user account SID S-1-5-21-1622753870... via \$RECYCLE.BIN artifacts[cite: 4]. However, disk forensics alone cannot establish physical keyboard attribution or subjective human intent without secondary domain audit logs or physical security controls.

12. Limitations & Recommendations

Limitations:

- **Directory Structural Unallocation:** Unallocated directory metadata for My pic, Our pic, and ADDS Project prevents direct path reconstruction in FTK Imager, requiring carving in Autopsy[cite: 1, 3, 5].
- **Attribution Scope:** Artifacts confirm user account SID S-1-5-21-1622753870... but cannot independently verify physical user presence.

Recommendations:

1. Enable object access auditing (Windows Event ID 4660/4663) for corporate directories.
2. Enforce centralized, write-once logging for endpoint activity.

13. Chain of Custody & Investigation Log

Chain of Custody Table

Field	Information
Case ID	AIT-DF-2026-009
Evidence ID	EV-01
Evidence Description	Forensic Disk Image (Evidence.E01)

Field	Information
Received From	Apex Integrated Technologies Ltd.
Received By	Ogooluwa Afolayan James
Date/Time Received	September 24, 2026 @ 09:00 WAT
Storage Location	Digital Forensics Secure Storage

Investigation Activity Log

Date/Time (WAT)	Tool	Action Executed	Result / Evidence Reference
2026-09-24 09:30	dd / Terminal	Calculated SHA-256 and MD5 hashes	Verification matched source values. DOCX
2026-09-24 10:15	FTK Imager 8.3	Mounted Evidence.E01, inspected [root]	Identified deleted SSH, Corel, corel draw work. PNG
2026-09-24 11:00	FTK Imager 8.3	Inspected \$RECYCLE.BIN\S-1-5-21...	Located desktop.ini (11:42:43 WAT)[cite: 4].
2026-09-24 11:30	FTK Imager 8.3	Inspected [unallocated space]	Cataloged raw sector chunks (0001929+)[cite: 3].
2026-09-24 12:40	Autopsy 4.23.1	Processed data source & filesystem ingest	Identified 1,162 deleted files (.cpt assets). PNG

Date/Time (WAT)	Tool	Action Executed	Result / Evidence Reference
2026-09-27 20:00	Autopsy / CLI	Extracted files & calculated output hashes	Generated hashes.txt for evidence folder. DOCX